

20260723.md: Check-in with Charleston

Project status

gLike

- Conda environment set up
- **IMPORTANT**
 - Replace

```
from setuptools import dist, setup, Extension

# bootstrap numpy; can we workaround this?

dist.Distribution().fetch_build_eggs(["numpy>=1.

# should be fine now

import numpy as np
```

- With:

```
from setuptools import setup, Extension
import numpy as np
```

- `fetch_build_eggs` is deprecated
- Installation works. [image]
- Progress-wise: I have run through the [tutorial](#) and am starting to look into the code.

- Specifically of interest is the `scipy.optimize.minimize`-like `glike.maximize` functionality. I dug around a lot in this particular `scipy.optimize` function during my PhD
- Off of the top of my head I can try implementing a few optimization strategies (to prioritize computational efficiency, global minima/maxima id, specific conditions):
 - Constrained optimization in which we ensure that parameter constraints are satisfied, e.g., one parameter larger or smaller than another, or that parameters sum to a specific value
 - Brute-force grid-based optimization, which should be better at finding global maxima with a trade off of efficiency
 - L-BFGS-B method (limited memory bfgs), which will work better when starting with parameters close to the true (global) optimum. Different starting positions are known to "burrow" to a singular optimum with this strategy. Worse when the initial guess is worse.
 - Basic idea is optimize based on matrix of second derivatives but only store like 5 matrices in memory
 - Direct search / Nelder-Mead (first-derivative) approach, which will work when far from any minimum; yielding a more robust search (for finding global minima), but much slower *search* and moderately faster *computation* due to lack of gradient information.
- Could even try a combination of something like Fmin (direct search) and L-BFGS-B, trading off speed and efficiency interchangeably to get over "dips" in the likelihood surface while gliding over the "easy to search" parts

Notes for Jon

- Command-line for login (must be on VPN): `ssh mahj@10.72.9.14` (IP for `discovery2` CARC server works for me)
- Command-line for interactive node: `salloc --time=2:00:00`

```
--cpus-per-task=8 --mem=16G --account=mahj_1946
```

- Computing workspace: /home1/mahj
 - **ASK CHARLESTON ABOUT THIS**
 - Myquota indicates that I have 100GB in this workspace, as well as a scratch with 10TB, and access to project2/chia657_28 (almost full), project2/haiman_625, and project2/minjuic_62
 - [image]
- .bashrc can be edited from home workspace.
 - module purge
 - module load conda
 - conda init bash
 - source ~/.bashrc
 - It doesn't like when I try and edit from nano, which is annoying (segmentation fault)
- Full installation sequence:
 - conda create -n glike python numpy=1.17.5 scipy pandas matplotlib tskit tsinfer msprime tqdm pip
 - conda activate glike
 - git clone <https://github.com/Ephraim-usc/glike.git>
 - cd glike
 - **Make Changes to setup.py**
 - python -m pip install --no-build-isolation -e .

Native Hawaiian Data

- I can confirm access to the data.
- [image]
- I have recieved detailed for SFTP transfer to the Michigan Imputation panel
 - User name: sftp-imp-refpanels-xfer
 - Password: 8gjF-jZxA-A26L-wkYE
 - Host: share.sph.umich.edu
 - Port: 2222
- **Check-in with Charleston**
 - I can go ahead and start transferring this via lftp and sftp
 - put
 - I'm pretty sure there's two dedicated transfer nodes on CARC for this. Looking into it

Standards for joint multi-sample variant calling

- I have recieved instructions from Xinran regarding how to start up a Terra account, but haven't yet done so